

WYNYARD, Australia

WMO: 959570

Lat: 40.997S

Lon: 145.729E

Elev: 12

StdP: 101.18

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.1	0.8	-1.2	3.4	4.9	-0.4	3.7	4.7	12.3	11.2	11.3	10.8	3.0	280	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.7	24.3	17.2	22.7	17.1	21.5	16.5	19.4	21.9	18.6	21.0	17.8	20.2	5.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	20.5	17.6	12.6	20.1	16.8	12.0	19.4	55.5	22.0	53.0	21.0	50.5	20.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	10.1	9.1	-1.9	29.2	0.6	2.5	-2.3	30.9	-2.7	32.4	-3.0	33.8	-3.4	35.6		
			-1.8	20.8	0.8	1.5	-2.4	21.9	-2.8	22.7	-3.3	23.6	-3.9	24.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	16.2	16.5	14.8	12.3	10.2	8.7	8.1	8.4	9.4	10.8	12.9	14.4
	DBStd	3.76	2.61	2.54	2.68	2.54	2.28	2.30	2.14	2.10	2.06	2.20	2.49	2.44
	HDD10.0	269	0	0	1	8	26	53	67	59	36	17	4	1
	HDD18.3	2384	76	61	114	181	252	289	317	308	269	232	163	123
	CDD10.0	952	192	183	149	77	32	13	8	9	17	42	92	138
	CDD18.3	26	10	9	3	0	0	0	0	0	0	0	1	2
	CDH23.3	109	55	26	8	1	0	0	0	0	0	0	6	14
	CDH26.7	17	12	3	1	0	0	0	0	0	0	0	1	1
(14) Wind	WSAvg	4.5	5.0	4.7	4.5	4.1	4.1	3.8	4.2	4.2	4.6	4.7	5.0	5.1
(15) Precipitation	PrecAvg	959	47	48	41	65	86	109	141	105	110	84	67	56
	PrecMax	1351	166	144	127	174	205	191	249	194	192	142	113	154
	PrecMin	783	6	4	4	13	18	38	31	58	36	31	15	10
	PrecStd	152	35	37	31	41	53	45	59	39	45	29	25	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	26.0	24.3	20.9	17.7	15.7	14.7	15.5	17.2	20.6	23.6	25.4
		MCWB	17.8	17.9	18.6	15.7	14.1	13.1	11.7	11.1	12.2	14.0	15.9	16.4
	2%	DB	24.4	23.6	22.2	19.2	16.5	14.7	13.7	14.3	15.6	18.1	20.5	22.2
		MCWB	17.0	17.6	17.2	15.4	13.7	12.3	11.2	11.0	11.6	12.8	14.8	16.1
	5%	DB	22.6	22.4	20.9	18.1	15.7	13.9	13.1	13.5	14.6	16.6	19.1	20.6
		MCWB	17.0	17.2	16.8	14.8	13.2	11.8	10.9	10.7	11.3	12.2	14.3	15.2
	10%	DB	21.2	21.2	19.8	17.3	15.0	13.3	12.5	12.7	13.7	15.5	17.8	19.3
		MCWB	16.2	16.8	16.1	14.3	12.8	11.3	10.4	10.4	10.9	11.9	13.8	14.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	20.4	19.9	17.5	15.6	14.1	13.0	12.9	13.8	15.3	17.7	18.9
		MCDB	24.6	22.6	22.4	19.0	16.3	15.0	13.6	13.8	15.1	17.8	20.2	22.0
	2%	WB	19.2	19.4	18.8	16.6	14.7	13.0	12.0	12.0	12.9	14.4	16.5	17.6
		MCDB	21.8	21.7	20.9	18.2	15.8	14.0	13.0	13.2	14.4	16.4	19.0	20.1
	5%	WB	18.3	18.7	17.8	15.8	13.9	12.3	11.3	11.4	12.1	13.5	15.4	16.8
		MCDB	20.8	21.0	19.9	17.7	15.2	13.6	12.6	12.8	13.8	15.5	17.9	19.1
	10%	WB	17.3	17.9	16.9	14.8	13.1	11.6	10.8	10.8	11.4	12.7	14.5	15.9
		MCDB	20.0	20.3	19.2	16.9	14.7	13.0	12.2	12.4	13.1	14.9	17.1	18.5
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	8.7	9.2	9.2	8.7	8.4	8.1	8.3	8.4	8.9	8.7	9.1
		MCDBR	11.6	10.0	9.8	9.7	8.9	8.0	7.9	8.6	9.4	10.4	10.7	11.3
		MCWBR	6.2	5.6	6.1	6.4	6.5	6.0	5.8	6.1	6.1	6.3	6.3	6.4
	5% WB	MCDBR	8.5	7.3	8.0	8.2	7.4	6.7	6.5	7.2	7.7	8.7	8.3	8.4
		MCWBR	5.8	5.0	6.0	6.2	6.0	5.4	5.3	5.7	5.8	6.2	5.7	5.7
(40) Clear-Sky Solar Irradiance	taub		0.355	0.346	0.334	0.326	0.315	0.307	0.308	0.324	0.348	0.343	0.343	0.349
	taud		2.498	2.538	2.573	2.571	2.580	2.587	2.588	2.545	2.462	2.492	2.515	2.505
	Ebn at Noon		973	956	925	863	805	778	805	852	893	949	979	984
	Edh at Noon		112	102	91	78	66	60	64	79	99	106	109	112
(44) All-Sky Solar Radiation	RadAvg		6.73	5.84	4.40	3.00	1.92	1.53	1.66	2.45	3.57	5.09	6.14	6.63
	RadStd		0.43	0.29	0.21	0.19	0.14	0.14	0.10	0.20	0.25	0.30	0.40	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.46	N/A	N/A	+0.90	+0.51	+0.49	N/A	-136	+128	+13
Regional (18 neighbors)	+0.27	N/A	N/A	+0.65	+0.41	+0.45	N/A	-80	+90	+11

Nomenclature: See separate page